WEEK 2 · DAY 2 · OVERVIEW

Day Counts and Accrued Interest



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ABOUT 45 MIN WITH EXERCISES / 8 SECTIONS / 4 EXERCISES

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The Desk Lead's Note

We buy 1,000,000 of Quorvane Energy, settling Monday 30 November, two months into its coupon period. The seller wants paying for the interest it earned: how much per 100 and in dollars, and why does a calendar give another number? Put the day counts and accrued interest in bondmath, tested against Excel. An assistant has offered to tidy that code; read its change before you take it. Last, show me the holdings file's accrued column is right for all 200 bonds.

Your Route Through Day 2

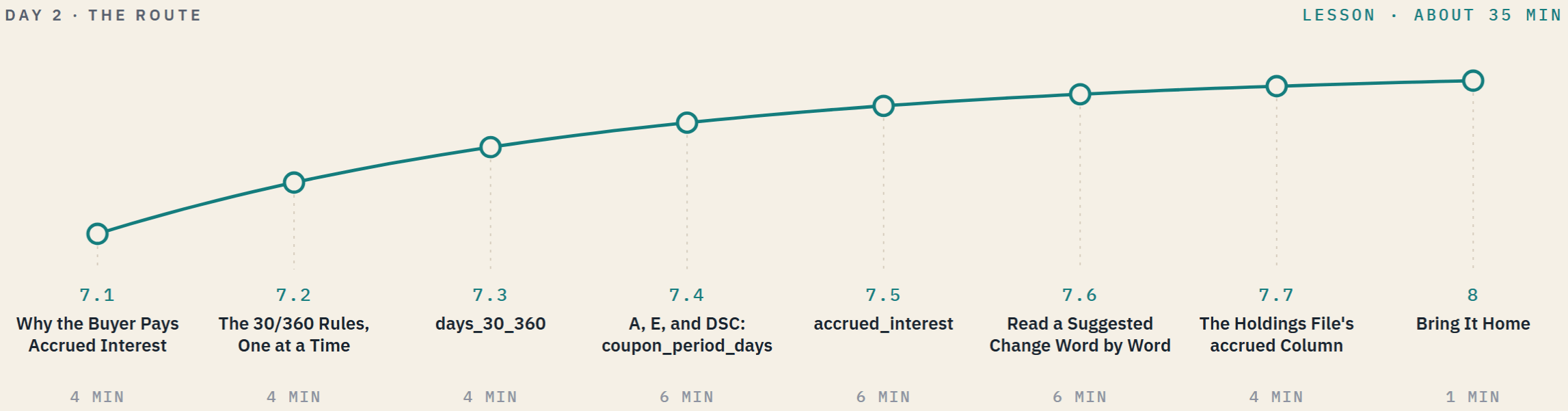


FIGURE 1 Eight sections, in order, with about how long each takes.

Time. About 35 minutes for the lesson, then about 12 for the four exercises.

7.1 Why the Buyer Pays Accrued Interest

COUNT CALENDAR DAYS

```
previous_coupon = date(2026, 9, 30)
next_coupon = date(2027, 3, 31)
days_held = (settlement - previous_coupon).days
days_in_period = (next_coupon - previous_coupon).days
accrued_actual = 5.25 / 2 * days_held / days_in_period
print(f'{days_held} of {days_in_period} days: accrued {accrued_actual:.6f} per 100')
```

```
61 of 182 days: accrued 0.879808 per 100
```

THE FOUR BONDS

	TICKER	MATURITY
0	QVNE	2031-09-30
1	DNMR	2033-09-30
2	PLWK	2029-09-30
3	HLVF	2036-09-30

accrued = coupon_pct / frequency x A / E

7.2 The 30/360 Rules, One at a Time

US 30/360: four rules move D1 or D2, then the formula counts

```
Days = 360 x (Y2 - Y1) + 30 x (M2 - M1) + (D2 - D1)
```

Rule	Case	Day numbers	Days
1 start is the last day of February: D1 = 30	28 Feb 2027 to 1 Mar 2027	D1 28 to 30	**3 to 1**
2 start and end both last of February: D2 = 30	28 Feb 2027 to 29 Feb 2028	D2 29 to 30	**359 to 360**
3 D1 is 31: D1 = 30	31 Aug 2026 to 30 Sep 2026	D1 31 to 30	**29 to 30**
4 D2 is 31 and the start date's day is 30 or 31: D2 = 30	30 Jul 2026 to 31 Oct 2026	D2 31 to 30	**91 to 90**

Rule 4 needs a start on the 30th or 31st: 15 Oct to 31 Oct 2026 keeps D2 = 31, so 16 days.

FIGURE 3 Four rules change a 30/360 count: 28 Feb to 1 Mar 2027 goes from 3 days to 1, and 30 Jul to 31 Oct 2026 from 91 to 90.

THE FORMULA WITH NO RULES

```
2026-08-16 to 2026-09-30: 44 days
2026-08-31 to 2026-09-30: 29 days
```

EXCEL SIDE BY SIDE

JOB	RESULT
The formula, start in A1 and end in B1	44, then 29
Rule 3 in the sheet	30

7.3 days_30_360

THE FOUR RULES

```
    if start_is_feb_end:
        d1 = 30
    if start_is_feb_end and end_is_feb_end:
        d2 = 30
    if start.day == 31:
        d1 = 30
    if end.day == 31 and start.day >= 30:
        d2 = 30
```

THREE FEBRUARY MONTH ENDS

```
2027-02-28 to 2028-02-29 360
2026-02-28 to 2026-10-31 241
2027-02-28 to 2027-02-28 0
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Rule 2	`=YEARFRAC(DATE(2027,2,28),DATE(2028,2,29),0)*360`, then `=DAYS360(DATE(2027,2,28),DATE(2028,2,29))`	360; DAYS360 359
Rule 4 after a February start	`=YEARFRAC(DATE(2026,2,28),DATE(2026,10,31),0)*360`, then `DAYS360`	241; DAYS360 240
The same date twice	`=DAYS360(DATE(2027,2,28),DATE(2027,2,28))`	0; DAYS360 -2

`days_30_360` **counts as** `COUPDAYBS` **and** `YEARFRAC(..., 0)` **do;** `DAYS360` **does not, on February month ends.**

7.4 A, E, and DSC: coupon_period_days

QUORVANE ON BOTH BASES

```
30/360 (60, 180, 120)
ACT/ACT (61, 182, 121)
Excel, basis 0: 60 180 120
```

WHERE COUPDAYSNC IS NOT E - A

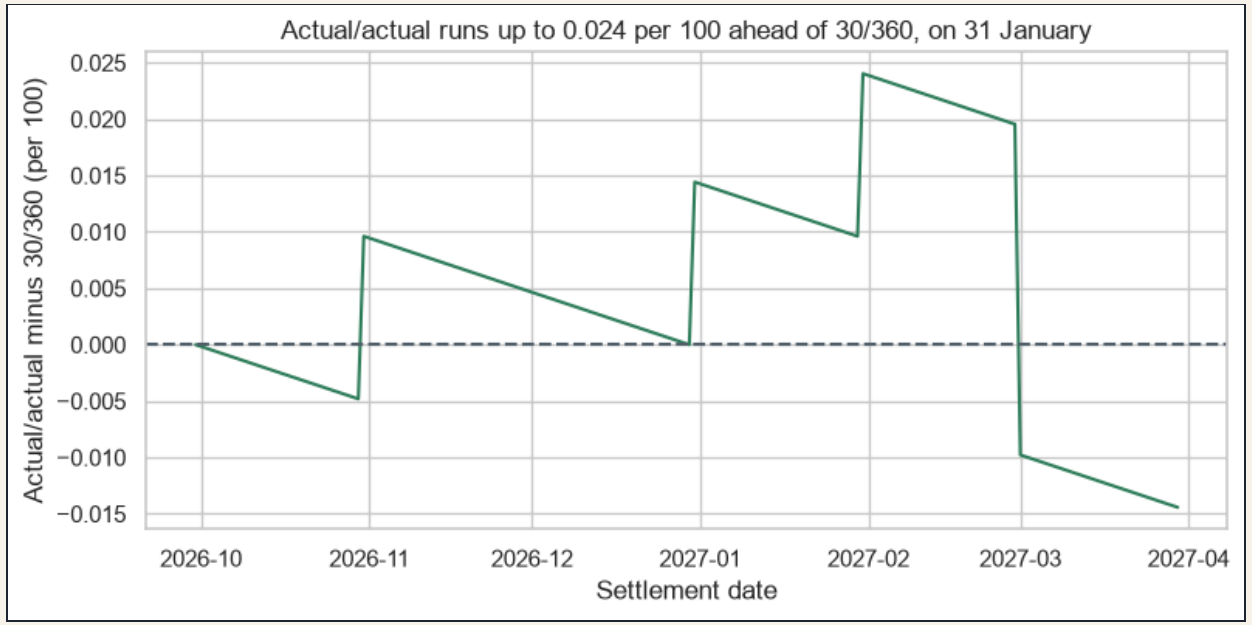
```
278 basis 0 rows; 7 where COUPDAYSNC is not E - A
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
A	`=COUPDAYBS(DATE(2026,11,30),DATE(2031,9,30),2,0)`	60
E	`=COUPDAYS(DATE(2026,11,30),DATE(2031,9,30),2,0)`	180
DSC	`=COUPDAYSNC(DATE(2026,11,30),DATE(2031,9,30),2,0)`; `COUPDAYBS(...)+COUPDAYSNC(...)`	120; 180

A, E, and DSC are `COUPDAYBS`, `COUPDAYS`, and the E - A that `PRICE` uses, on every reference row.

7.5 accrued_interest



FOUR BONDS AGAINST EXCEL

	TICKER	PYTHON	EXCEL
0	QVNE	0.875000	0.875000
1	DNMR	1.020833	1.020833
2	PLWK	0.791667	0.791667
3	HLVF	1.250000	1.250000

ACCRINT, THE SECOND ROUTE

```
ACCRINT blank (#NUM! in Excel): 21  different: 9
```

`accrued_interest` **is Excel's accrued formula on all 306 rows, both bases: the desk pays the seller 8,750.00 dollars.**

7.6 Read a Suggested Change Word by Word

Reading git diff --word-diff: only the words that changed are marked

```
{+# a year has 365 days+}
days_in_period = [-360-]{+365+} // frequency
days_to_next = days_in_period - days_accrued
```

`{+ +}` added

`[- -]` removed

no brackets: untouched

One number changed inside one line: 360 became 365, and a comment came with it.

FIGURE 5 Git's word diff marks 360 as removed and 365 as added in the line days_in_period = 360 // frequency.

WORD BY WORD

```
!git -C "{work}" diff --word-diff bondmath.py
```

THROW IT AWAY

```
!git -C "{work}" restore bondmath.py
Accrued again: 0.875000
```

Read every suggested change word by word and let the tests from Excel judge it: 360 is the convention, not a typo.

7.7 The Holdings File's accrued Column

RECOMPUTE ALL 200

```
holdings = pd.read_csv('holdings.csv')
holdings['accrued_python'] = [
    bondmath.accrued_interest(date.fromisoformat(a), date.fromisoformat(m), c)
    for a, m, c in zip(holdings['as_of_date'], holdings['maturity'], holdings['coupon'])]
holdings['gap'] = (holdings['accrued_python'] - holdings['accrued']).abs()
print(f"Largest gap: {holdings['gap'].max():.10f} per 100")
```

CHECK THE LIMIT

```
assert holdings['gap'].max() <= 0.0005 + 1e-9, 'a bond is off by more than rounding'
print('Bonds at a rounding tie:', ((holdings['gap'] - 0.0005).abs() < 1e-9).sum())
```

THE FILE'S ROUNDING

Rounding to 3 decimals moves a number by at most 0.0005, half of the last decimal.

Bring It Home

COPY, TEST, COMMIT

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_07_k4w7q2xr"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py", "$work\.gitignore" .
python -m pytest -q
git diff --stat
git add bondmath.py test_bondmath.py .gitignore
git commit -m "Notebook 07: days_30_360, coupon_period_days, and accrued_interest"
```

Keep the habit: read any suggested edit with `git diff --word-diff`, run `python -m pytest -q`, and `git restore` it if a test fails.

You Can Now

- Compute accrued interest as one coupon times A / E, and say why the buyer pays it.
- Count days on US 30/360 with **`days_30_360`** or **`YEARFRAC(..., 0)`**, and say where **`DAYS360`** differs.
- Count a period on actual/actual, with E as **`COUPNCD - COUPPCD`**.
- Return A, E, and DSC with **`coupon_period_days`**, tied to Excel's **`COUP`** functions.
- Compute accrued with **`accrued_interest`**, and say when **`ACCRINT`** differs.
- Read a suggested change with **`git diff --word-diff`**, judge it with pytest, and discard it with **`git restore`**.
- Explain a gap of exactly half a file's last decimal.

NEXT
WEEK 2 · DAY 3

Clean and Dirty Price

Price a bond between coupons: the quote, the invoice, and a price and a yield for any settlement date, tied to Excel's PRICE and YIELD.